Heriot Watt University,
Department of Mathematics

Dr J Gaunt &
Dr J Tan

Assessed Coursework 1

Module F18CD 2025–2026

Handed out: Friday 14th November 2025 (2PM UK time)

Due: Friday 10th October by:

- **Edinburgh: 2PM**
- **Malaysia: 9PM**
- **Dubai: 5PM**
- You must submit your work via the Gradescope submission link available on assignment page in Canvas.
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 2 questions, totalling 20 marks.
- Unless stated otherwise, you may utilise any theorems proven in the lecture notes. Make sure to state explicitly which theorems you use from the notes in your answers.

Question 1

(a) Use the Monotone Convergence Theorem to prove the comparison test for series.

(b) Prove that if $\sum_{k=1}^{\infty} a_k$ converges absolutely, then $\sum_{k=1}^{\infty} a_k^2$ also converges absolutely.

(c) Determine whether the converse of (b) is true or false. Provide a proof if it is true, otherwise, provide a counterexample.

[8 marks]

Question 2

(a) Prove that $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k(\ln(1+k))^m}$ is convergent if and only if $m > 1$. (Hint: Use Cauchy's condensation test)

(b) Let (F_n) be a sequence defined recursively by $F_{n+2} = F_{n+1} + F_n$ with $F_1 = F_2 = 1$. Now consider the series defined by

$$\sum_{k=1}^{\infty} \frac{1}{F_k F_{k+2}}.$$

Prove that the series is convergent and find what it converges to. Hint: Prove that

$$\frac{1}{F_k F_{k+2}} = \frac{1}{F_k F_{k+1}} - \frac{1}{F_{k+1} F_{k+2}}.$$

(c) Use part (b), or otherwise, to find the sum of

$$\sum_{k=1}^{\infty} \frac{1}{F_k F_{k+4}}.$$

[12 marks]